

Project 4

Digit Classification using RBF Network

Introduction

In this computer project, you will design a radial bases function (RBF) neural network followed by one layer regular neural network for decimal digit classification using the K-Means and backpropagation algorithms. You should use the MNIST data set (see Project 3 details) that consists of 60,000 training images and 10,000 test images. The training set should only be used for training, and the test set should only be used for testing. Specifically, You will design a decimal digit classifier based on the hybrid Radial Basis Function (RBF) Neural Network, k-means clustering, and the backpropagation algorithms.

1. K-Means Clustering (100 Clusters)

A. Use only the data in `train-images` file (treating it as unlabeled) and cluster the training images into 100 clusters using the k-means algorithm.

B. Run k-means algorithm five times with different (randomly chosen) initialization for the clusters centers and choose the best result which gives minimum total distortion.

C. Print (output) the centroids of the best clustering that you obtained as program output.

D. Save the centroids of the best clustering as rows in an excel sheet named `centers_100`.

E. Save your code in a file named `kmeans_100`.

2. RBF Neural Network (Based on 100 clusters)

Create a separate file named `RBF_100` that performs the following steps:

A. Read the excel file `centers_100` (which contains 100 rows representing the centroids of the best clusters generated by the k-means algorithm).

B. Use these centroids of the clusters as the mean values for the RBF neurons.

C. Assign the same variance to all RBF neurons and compute it as discussed in class and lecture notes, show your calculations.

D. The RBF network output is the input to the output layer which has 10 neurons that represent the predicted decimal symbol in the input image.

E. Train the RBF-Output layer using only `train-images` and `train-labels`:
   - Present one example at a time.
   - Update the weights using backpropagation algorithm with a learning rate of 0.2

F. Experiment with different learning rates or settings to minimize error while avoiding overfitting. Document any hyperparameter changes at the beginning of your code as comments and in your report.

G. Plot the mean squared error (MSE) for the training data and save the plot as `TrainMSE_100`.

H. Evaluate your classifier using `t10k-images` and `t10k-labels`, and plot the miscalculation on the test data (accuracy) and save it as `Accuracy_100`.

3. Repeat for 10 and 500 Clusters

Repeat Steps 1 and 2 for 100 and 500 clusters. Use the following file names please:

- **For 10 clusters:**
  - `centers_10`, `kmeans_10`, `RBF_10`, `TrainMSE_10`, `Accuracy_10`
- **For 100 clusters:**
  - `centers_500`, `kmeans_500`, `RBF_500`, `TrainMSE_500`, `Accuracy_500`

4. Submissions

Include the following in one compressed file and name it as your last name_P4.

1. Submit a report summarizing your approach and results. Include all figures (`TrainMSE`, `Accuracy`) for each case and describe:
   - How did you select hyperparameters?
   - How did you avoid or detect overfitting?
2. All figures (you should have 6 figures).
3. All programs (you should have 6 diffident programs)

5. Code Quality

- All parameters in your code should be assigned at the beginning of the code.
- Comment your code thoroughly so the grader can easily follow the logic.
- Each file should be clean, organized, and reproducible.

6. Grading

Project evaluation will be based on:

- Code quality and documentation.
- Classifier performance.
- Quality of analysis and results.